Lecture 17 – Formulas for the determinant

Elim: $A \longrightarrow U$

$\rightsquigarrow \det A = \pm \det U$

$= \pm$ product of pivots.

Example:

$$A = \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 \end{bmatrix}$$

$$\rightsquigarrow U = \begin{bmatrix} 2 & -1 & & & \\ 0 & \frac{3}{2} & -1 & & \\ & 0 & \frac{4}{3} & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & 0 & \frac{n+1}{n} \end{bmatrix}$$

$\rightsquigarrow \det = n+1$

Rule 3: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix}$

$$= \cancel{\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix}}^{\,0} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \cancel{\begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}}^{\,0}$$

$$= ad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + bc \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = ad - bc.$$

$n^n = 2^2$ easy determinants: one entry in each row

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & & \\ & a_{22} & \\ & & a_{33} \end{vmatrix} + \begin{vmatrix} & a_{21} & \\ & & a_{23} \\ a_{31} & & \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ a_{21} & & \\ & a_{32} & \end{vmatrix} +$$

$$+ \begin{vmatrix} & & a_{13} \\ & a_{22} & \\ a_{31} & & \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ a_{21} & & \\ & & a_{33} \end{vmatrix}$$

$$+ \begin{vmatrix} a_{11} & & \\ & & a_{23} \\ & a_{32} & \end{vmatrix} + \underbrace{\begin{vmatrix} a_{11} & & \\ a_{21} & & \\ & & a_{33} \end{vmatrix}}_{\text{Column of zeros.}}^{\,0} + \cdots^{\,0}$$

Only $n!$ of the easy determinants will be nonzero: entries in different columns.

$$\begin{vmatrix} & a_{12} & \\ a_{21} & & \\ & & a_{31} \end{vmatrix} = a_{21}a_{12}a_{31} \begin{vmatrix} & 1 & \\ 1 & & \\ & & 1 \end{vmatrix}$$

Permutation matrix: rows of the identity matrix, in some order.

Big Formula:

$$\boxed{\det A = \sum_{\substack{P \text{ perm.} \\ \text{matrix}}} a_{1\alpha} a_{2\beta} \cdots a_{n\gamma} \cdot \det P}$$

$$\det P = \begin{Bmatrix} +1 \\ -1 \end{Bmatrix} \Leftrightarrow \begin{Bmatrix} \text{even} \\ \text{odd} \end{Bmatrix} \# \text{ of swaps.}$$

Example:

$$\begin{vmatrix} & & 1 \\ 1 & & \\ & 1 & \end{vmatrix} = -1, \quad \begin{vmatrix} & 1 & \\ & & 1 \\ 1 & & \end{vmatrix} = +1$$

Checking rule 3

Terms involving a_{11}:

$a_{11} C_{11}$

"Cofactor"

$$C_{11} = \sum_{P} a_{2\beta} \cdots a_{n\gamma} \cdot \det P$$

= determinant of submatrix.

$$\boxed{\det A = a_{11} C_{11} + a_{12} C_{12} + \cdots + a_{1n} C_{1n}}$$

$$\det A = a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{23}a_{32} - a_{21}a_{33}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

$$= \begin{vmatrix} a_{11} & & \\ & a_{22} & a_{23} \\ & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} & a_{22} & \\ a_{21} & & a_{23} \\ a_{31} & & a_{33} \end{vmatrix} + \begin{vmatrix} & & a_{33} \\ a_{21} & a_{22} & \\ a_{31} & a_{32} & \end{vmatrix}$$

$$C_{1j} = (-1)^{1+j} \det M_{1j}$$

M_{1j} = A without row 1 and column j.

In general:

$$\det A = a_{i1} C_{i1} + \cdots + a_{in} C_{in}$$

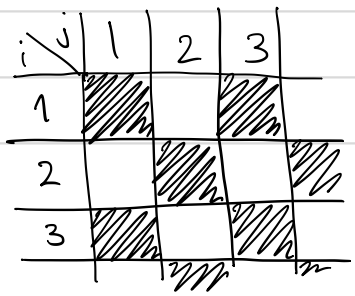
$$C_{ij} = (-1)^{i+j} \det M_{ij}$$

M_{ij} = A without row i and column j.

Sign: $(-1)^{i+j}$

$\blacksquare = +1$

$\square = -1.$



Example:

$$A_4 = \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix}$$

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 2 & -1 & \\ -1 & 2 & -1 \\ & -1 & 2 \end{vmatrix} = \det A_3$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} -1 & -1 & \\ & 2 & -1 \\ & -1 & 2 \end{vmatrix} = \det A_2$$

$$\det A_4 = 2 \cdot \det A_3 - \det A_2$$

$$\boxed{\det A_n = 2 \cdot \det A_{n-1} - \det A_{n-2}}$$

$$\Rightarrow \det A_n = n+1$$

$$n+1 \overset{\checkmark}{=} 2(n-1+1) - (n-2+1)$$

Application: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{\det A} \underbrace{\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}_{\begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix}}$

In general:

$$\boxed{A^{-1} = \frac{1}{\det A} C^T}$$